IBM Z DATATHON 2026 · REAL-TIME AI FOR CRITICAL DECISIONS

PayeeShield

Stop the payment before the money’s gone.

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ALLOW · WARN · BLOCK

WHERE THIS STARTED

A scammer gave our class tutor a number to pay. Our tutor sent ₹1. It went through. The number was “out of service”.

STEP 1

UPI checks the SIM **once**, by SMS, when the account is registered.

STEP 2

The scammer switches off or throws away the SIM.

STEP 3

The number **keeps receiving money**. Nobody checks again.

The problem is big, and mostly invisible

56.5%

of all bank frauds reported in FY25 were digital payment frauds.

51%

of UPI fraud victims never filed a complaint anywhere.

98.5%

of fraud value comes from payments above ₹10,000.

UPI is instant, so once the money moves, recovery is hard. **Prevention has to happen before the payer confirms.**

Sources: RBI Annual Report 2024–25 · LocalCircles survey, 2025 (32,000+ respondents) · RBI discussion paper on digital payment safeguards

The gap: payment and telecom never talk again

12 months

before NPCI removes a UPI ID whose number is inactive. Active scam numbers are never “inactive”.

90 days

before a telecom can give a disconnected number to someone new.

Minutes

to create a new UPI ID and start collecting scam payments.

The fraud signal lives on the **receiving side** — and it changes by the hour, not by the year.

Sources: NPCI circular on inactive UPI IDs (Nov 2023, 12-month rule 2025) · TRAI number reassignment rule · mFilterIt mule report, Mar 2026

What exists today, and where PayeeShield fits

Tool	What it watches	What it misses
NPCI real-time warnings	The payer: amount, frequency, device	Whether the receiver is alive and legitimate
RBI MuleHunter.AI	Mule accounts, inside each bank	Live telecom status at the moment of payment
NPCI 12-month delinking	Long-dead numbers	Active scam numbers
PayeeShield	The payee, live: SIM, UPI ID age, senders, outflow	Plus why — in plain words, to the payer

Sources: NPCI and RBI announcements, 2025–26 · Supreme Court order on digital arrest scams, 4 Aug 2026

How it works: ten signals, one decision, in milliseconds

Payee signals

- SIM reachable on the network?
- SIM age and swaps in 90 days
- UPI ID age
- Unique senders in 24 hours
- Minutes before money moves out

Payment context

- Amount and hour
- First payment to this payee?
- Payer on a call while paying?

Gradient-boosted model

Risk score 0–100, plus rule-based reason codes anyone can read.

Decision

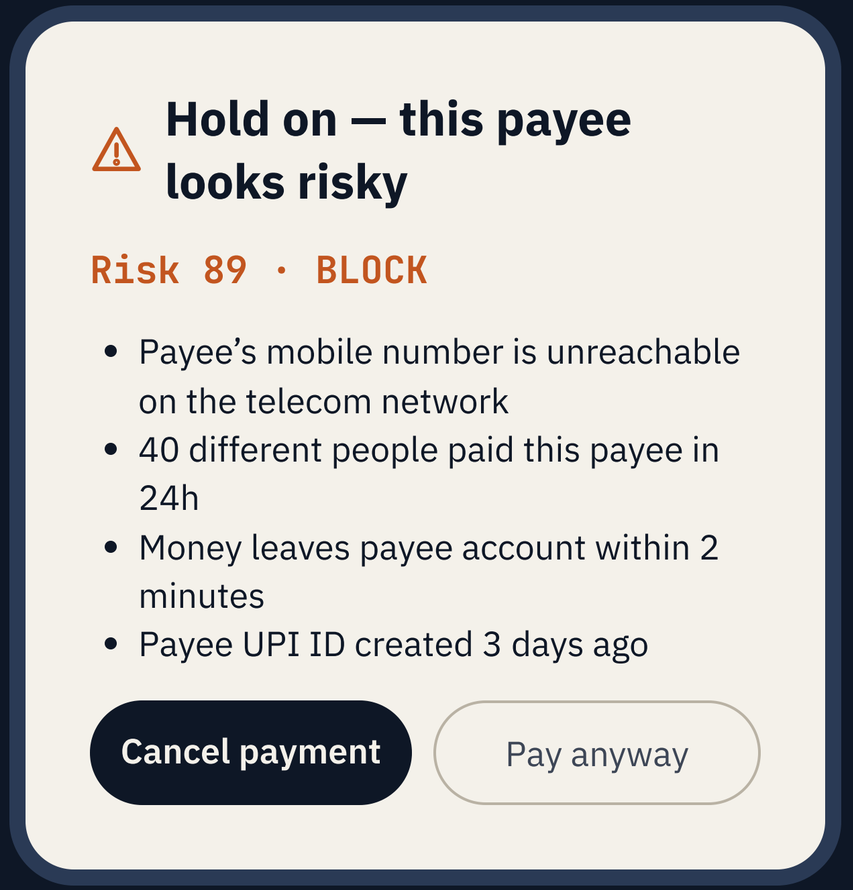
ALLOW below 40
WARN 40–74: pause and show reasons
BLOCK 75+

Payer app → UPI switch → **PayeeShield on IBM Z** → decision and reasons back to the payer app, before the PIN is accepted.

THE ₹1 STORY, REPLAYED

What the payer sees instead of “Payment successful”

Same payment our tutor made. Our trained model scores it **89 out of 100 — BLOCK** — and says why, in about 3 milliseconds.



Payee signals make the model much sharper

0.83

PR-AUC with payee signals, vs **0.48** using payment data alone

91%

of fraud caught at WARN or above

0.964

ROC-AUC, vs 0.894 for the payment-only baseline

~3 ms

median to score one payment on a standard CPU (6.5 ms p95)

Fraud pattern caught (at WARN)	Recall	Main signal
Dead-SIM mule	95%	SIM unreachable, many strangers paying
Digital arrest	100%	Large amount, new payee, payer on a call
Fake refund / QR	86%	New UPI ID, first-time payee

Synthetic test set: 10,000 payments, 3.1% fraud, 80/20 stratified split. SIM-swap mules: 100% recall.

What we know we haven’t solved yet

Synthetic data

No public dataset has payee telecom signals. Ours is built from reported fraud patterns; real bank data would change the numbers.

New shops look like mules

34% of brand-new legit shops get a WARN. That’s why WARN is a pause with reasons, not a block.

Needs a telecom feed

SIM status and swaps would come from telecom or DoT risk data shared with banks. We simulate it.

Not yet run on IBM Z

The prototype is plain Python, built to port to LinuxONE. Latency there is still to be measured.

Why IBM Z: score inside the payment, not after it

In the transaction path

On-chip AI acceleration on IBM Telum processors is built for scoring every transaction as it runs, not sampling.

Data stays put

Payment and telecom signals are scored where they already live — nothing sent to an outside AI service.

Built for UPI scale

UPI handles over 750 million payments a day. Banks already run their core systems on mainframes.

Path to deploy: same Python service on **LinuxONE**, model exported for the IBM Z AI stack.

Next: from prototype to pilot

01

Benchmark scoring latency on LinuxONE

02

Plug in telecom risk data and the I4C suspect registry

03

Pilot with a bank on real, labelled cases

04

Warnings in Tamil, Hindi and other languages

Our tutor’s ₹1 went through. The next person’s ₹1,50,000 shouldn’t.

PayeeShield · Team ACHILLES · Code and live demo linked in our submission